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Production of radical species by electron beam deposition in an ArF* lasing medium

G. M. Petrov, M. F. Wolford, Tz. B. Petrova, J. L. Giuliani, and S. P. Obenschain
Naval Research Laboratory, Plasma Physics Division, 4555 Overlook Ave., SW, Washington, DC 20375, USA

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The electron-beam-pumped ArF laser is a laser technology capable of providing very high lasing energies at the shortest wavelength ($\lambda = 193$ nm) among the rare gas halide lasers and therefore has the potential to be a superior driver for inertial fusion. The electron kinetics are rigorously treated by numerically solving the steady-state, spatially averaged electron Boltzmann equation in Ar-F_2 gas. The *e*-beam energy deposition and collisional reaction rates with electrons are calculated from the electron energy distribution function for a wide range of *e*-beam deposition powers ($P^{beam} = 10\,\mathrm{kW/cm^3}$–$3\,\mathrm{MW/cm^3}$) and fluorine concentrations ($x_{F_2} = 0.01 - 10\%$). The rates are reduced to a set of coefficients that are fitted with analytical formulas as a function of two universal parameters: P^{beam}/p and x_{F_2}, where p is the gas pressure. It is found that in the regime of high *e*-beam power deposition, the fluorine molecules are rapidly destroyed through dissociative attachment and neutral dissociation. The loss of F_2 over the duration of the beam is proportional to the *e*-beam energy deposition per unit volume, ε^{beam}, and follows $\Delta n_{F_2}(cm^{-3}) \cong 4 \times 10^{17} \varepsilon^{beam}(\mathrm{J/cm^3})$, in agreement with experimental data. The fluorine molecule conversion to other fluorine species, including atomic fluorine, is shown to have a very small effect on the index of refraction even at percent level concentrations. *Published by AIP Publishing*. https://doi.org/10.1063/1.4995224

I. INTRODUCTION

The krypton fluoride (KrF) laser has long been an attractive candidate for direct drive inertial confinement fusion (ICF).[1] It has shorter wavelength ($\lambda = 248$ nm) than the frequency tripled Nd:glass lasers ($\lambda = 351$ nm) used on many large laser fusion facilities and can provide more uniform illumination of targets. Its beam smoothing technology also provides the capability to zoom down the focal diameter to follow an imploding pellet. The deep ultra-violet (UV) light increases both the coupling efficiency to the target and the intensity thresholds for undesired laser-plasma-instability. The superior pellet-illumination would allow more symmetric implosions while the focal zooming further increases the coupling efficiency. In turn, the argon fluoride (ArF) laser could be superior to KrF as a driver for ICF. It has (i) a still shorter wavelength (193 nm vs. 248 nm) leading to yet better coupling efficiency between the laser light and the imploding pellet; (ii) innately broader frequency bandwidth that will improve the beam smoothing on target, which relies on controlled laser incoherence. The broader bandwidth could also further help mitigate laser plasma instabilities that depend on resonances between the laser and plasma waves. The ArF* excimer molecule also has characteristics that should increase the intrinsic efficiency and that are helpful in the construction of large, high-energy amplifiers. These include (i) smaller absorption cross section by fluorine molecules at this wavelength (1.5×10^{-21} cm^2),[2] which increases the efficiency and allows the utilization of larger volume amplifiers; and (ii) higher formation efficiency of the ArF* excimer, on the order of 15%–20%.[3,4] The feasibility of achieving higher wall plug efficiency at shorter wavelength makes the ArF laser a very attractive choice as a driver for ICF and in the longer term as a driver suitable for inertial fusion energy (IFE).

In order to become a potential driver for ICF, the laser system must possess high intrinsic efficiency. Small electron-beam (e-beam) pumped ArF lasers were studied experimentally in the 1970s and 1980s (Refs. 5–13) with output energies up to 92 J.[5] Similar parametric studies of the basic physics of an electron-beam-pumped ArF laser are being planned for the 30-cm aperture *Electra* facility at the US Naval Research Laboratory which was utilized earlier for KrF laser research.[14] Review of available data from experiments and simulations reveals a vast spread of conversion efficiencies, from ~1% to ~15% in experiments and from to 15% to 23% from theory and simulations. The wide data variation, particularly the experimental ones, cannot be easily explained. To do so, the ArF laser kinetics needs to be revisited. The starting point is the e-beam deposition kinetics, which is the primary objective of this paper.

The Ar-F_2 kinetics has been exhaustively investigated, primarily in the context of the KrF laser, which is the most studied among all excimer lasers. Since the majority of KrF lasers use a triple mixture of Ar, Kr, and F_2, the Ar-F_2 kinetics can be considered as a subset of the KrF laser kinetics.[10,15–19] Although gas kinetics models for the ArF laser exist, the focus was almost entirely on discharge plasmas.[20–23] Only a handful of models for e-beam driven plasmas exist.[3,4,7,24] However, for ICF applications, large volume lasers at near atmospheric pressure pumped by intense *e*-beams are the most promising in terms of scalability and energy output. The formation and quenching kinetics of the ArF* excimer was studied in the late 70s (Ref. 7) and mid-80s.[25] Since then, the most advanced gas kinetics model of the *e*-beam driven ArF laser was developed by Lee *et al.*,[4] which includes extensive plasma chemistry,

laser beam intensity amplification along the amplifier length, small signal gain, saturation intensity, and stage gain calculation. The electron collisional rates were, however, calculated using a less rigorous method: the electron energy distribution was modeled as a delta function at the electron beam energy superimposed on a Maxwell distribution. Since the electron kinetics is essential for the subsequent gas kinetic reactions, excimer formation and lasing, it is imperative to treat it self-consistently. As it will be demonstrated in Sec. III, critically important electron collisional rates cannot be calculated accurately using a Maxwell distribution. This task can be accomplished by solving the electron Boltzmann equation for the electron energy distribution function (EEDF) and has the advantage of using the EEDF in the entire energy region rather than decomposing it artificially as it was done in Ref. 4. The importance of the EEDF was recognized early on with energetic electrons (above the excitation threshold of the rare gas) being responsible for excitation and ionization, while bulk electrons dominate recombination, attachment, and quenching of excited states. By the mid- and late 1980s, it was implemented in studies of the KrF and ArF lasers.[24]

Since accurate collisional rates are the backbone of any numerical model, a comprehensive theoretical study of the energy deposition in an *e*-beam pumped amplifier and computation of collisional rates with electrons will greatly benefit future exploration of the ArF excimer laser. The purpose of this paper is twofold: (i) we wish to rigorously compute the collisional rates with electrons using the calculated EEDF and (ii) represent these rates in the form suitable for further species kinetics studies. The simulations conducted in this work cover a wide range of plasma parameters such as mean electron energy and density, production rates, and rate coefficients. The production rates are then used to calculate a set of slowly varying coefficients, among which are the energy per electron-ion-pair and excitation-to-ionization ratios. Details of the numerical procedure are outlined in Sec. II. The simulation results are presented in Sec. III as a function of two generalized parameters: normalized e-beam power per unit volume and F_2 concentration. In order to aid kinetic studies, the computed coefficients have been fitted with appropriate analytical functions and summarized in a table. The present study is limited in scope covering only reaction rates with beam electrons and their secondaries. However, within this study, we can use the calculated set of rates to address the problem of fluorine burn-up, which is essential for the effective operation of the ArF laser because burn-up has implications regarding the choice of initial gas composition. Since fluorine molecules represent only a small fraction of the total number of particles ($\sim$0.3%) and the power deposition density is rather large, it is conceivable that fluorine molecules would be consumed by *e*-beam reactions, rather than for lasing. Due to the low absorption of the 193 nm lasing line by F_2, there should be a minimal penalty for increasing the number density of F_2, if necessary. Therefore, in Sec. IV, we study the fluorine burnup and offer recommendations for counteracting its impact. We also investigate the effect of electron density and fluorine burnup on the index of refraction for the gain media. A summary of the results is given in Sec. V.

II. IMPLEMENTATION

The application of the present work is specific to high-power ($\sim$1 MW/cm^3) *e*-beam energy deposition in a large ($30 \times 30 \times 100$ cm^3) laser amplifier at atmospheric pressure (1–3 atm.) over a period of time on the order of hundreds of nanoseconds. The modeling approach adopted in this work has been reported elsewhere,[26,27] and only the main assumptions will be recalled, namely: (i) the electron beam is completely stopped by the gas; (ii) the energy deposition is spatially uniform; (iii) the EEDF relaxation time to a quasi-steady state is shorter than the beam duration; (iv) electric fields generated due to return currents have been neglected; and (v) only processes with ground state atoms and molecules are considered. The last assumption means that the concentration of reactive species formed by the e-beam and subsequent gas chemistry remains small compared to the initial concentration of Ar and F_2. The cross sectional sets of Ar and F_2 are the same as in our previous works,[26] except that the ionization cross section of F_2 is split into ionization to F_2^+ and dissociative ionization to F^+ and F. The only input parameters are the gas pressure, F_2 concentration, energy of the beam electrons, and *e*-beam power deposition per unit volume, P^{beam}. The electron density n_e is a computed quantity and explicitly enters the electron Boltzmann equation. It is determined by balancing ionization with dissociative recombination (superscript *dr*) and attachment (*att*) to fluorine molecules as follows:

$$k^{dr}_{Ar_2^+}(T_e)n_e^2 + k^{att}_{F_2} n_{F_2} n_e = R^{ion}_{tot}. \quad (1)$$

The total ionization rate in the right hand side of Eq. (1), attachment rate coefficient $k^{att}_{F_2}$ to F_2 molecules, and electron temperature T_e are calculated from the EEDF. Equation (1) is a rough estimate for n_e. Electrons are lost in a number of recombination processes, but only the leading one is retained. Recombination with atomic ions, $Ar^+ + e + M \rightarrow Ar^* + M$, with $M = Ar,e$ being the third particle, has a much smaller rate and has been neglected. A more detailed version of Eq. (1) includes excited states of argon (Ar*) and numerous reactions involving stepwise ionization, collisional deactivation to ground state, and destruction of Ar* with F_2 molecules. This issue can be addressed by detailed chemistry models, but in this study, the electron density enters only the non-linear Coulomb term in the electron Boltzmann equation. All processes participating in Eq. (1) are assumed to be "instantaneous," but it has to be recognized that each term may have its own time scale. There is an additional time lag between ionization and electron loss in attachment and recombination due to the finite time it takes to complete the electron degradation in energy space. The latter may take 20–40 ns,[24] depending on the initial beam energy and gas pressure. The dissociative recombination time, $\tau^{dr} = (k^{dr} n_e)^{-1} \simeq 10 - 20$ ns, is comparable. Both processes are fast and proceed on a time scale shorter than the typical *e*-beam pulse duration, which justify the assumptions made for Eq. (1).

The typical gas mixture in the Ar-F_2 amplifier is $\geq$99% Ar and $\leq$1% F_2. If one of the gases (argon) is dominant and the other (fluorine) is an admixture, the input parameters

$\{P^{beam}, n_{F_2}, \text{and } n_{Ar}\}$ can be reduced to a combination of two parameters, $\{\frac{P^{beam}}{n_{tot}}, \frac{n_{F_2}}{n_{tot}}\}$, where $n_{tot} = n_{Ar} + n_{F_2}$ is the total number density. This claim can be derived from the electron Boltzmann equation by observing that the left hand side is proportional to species densities, while the right hand side is proportional to the beam ionization rate, which, in turn, is proportional to P^{beam}. Upon dividing both sides by the total number density, we get the two universal parameters listed above. Strictly speaking, there is a third normalized parameter, n_e/n_{tot}, which introduces a slight non-linearity in the electron Boltzmann equation. Numerical simulations confirm that these scaling laws indeed hold and that only two normalized parameters need be used instead of three, namely, the reduced beam power, P^{beam}/p, and concentration of fluorine molecules, $x_{F_2} = n_{F_2}/n_{tot}$. This is in contrast to our previous study in which the EEDF was calculated in a triple mixture Ar-Kr-F_2 with approximately equal amount of Ar and Kr, for which such simplification was not possible.

After solving the electron Boltzmann equation for the EEDF, the collisional rates of all individual processes with electrons are calculated. These rates are then used to produce a set of slowly varying coefficients, described below. A widely used parameter in *e*-beam generated plasmas is the energy per electron-ion pair, which is defined as

$$W_{ei} = P^{beam}/(R^{ion}_{tot} e_0), \tag{2}$$

where $R^{ion}_{tot} = \sum_{\alpha=Ar,F_2} R_\alpha$ is the total ionization rate in the plasma, which is summed over the combined ionization rates $R_\alpha = R^{ion}_\alpha + R^{di}_\alpha + R^{Aug}_\alpha$ of species $\alpha =$ Ar, F_2. In Formula (2), the magnitude of the electron charge e_0 is required as a conversion factor from joules to electronvolts, in which W_{ei} is usually given. The superscripts *ion*, *di*, and *Aug* denote the type of ionization process: ionization, dissociative ionization, and Auger ionization, respectively. The total ionization rate is partitioned among all the species α. The combined ionization rate R_α of each species α is proportional to the species concentration x_α with proportionality coefficient γ_α, for which one can write

$$\gamma_\alpha = \frac{R_\alpha}{x_\alpha R^{ion}_{tot}}, \tag{3}$$

where $x_\alpha = n_\alpha/\sum_\alpha n_\alpha$ is the fractional number density of species α. The excitation-to-ionization ratios are calculated for each species α, individual level k, and type of process β according to

$$\eta^\beta_{\alpha,k} = R^\beta_{\alpha,k}/R_\alpha. \tag{4}$$

The numerator is the rate of the individual process (e.g., excitation to level k). The advantage of using the parameters W_{ei}, γ_α, and $\eta^\beta_{\alpha,k}$ is that they vary slowly in a wide range of input parameters. In Sec. III, reactions due to the *e*-beam and secondary electrons are formulated as a function of the combined input parameters $\{P^{beam}/p, x_{F_2}\}$ and fitted with appropriate formulas. Gas kinetic calculations can use these formulas to evaluate the rates of individual processes by inverting expressions in (2)–(4).

III. SIMULATION RESULTS

The calculated EEDFs as a function of electron energy u are plotted in Fig. 1 for various fluorine concentrations, x_{F_2}. The EEDFs are normalized according to $\int_0^\infty f(u)du = n_e$. For these results, the gas pressure and electron beam power density are fixed at the nominal values expected in the *Electra* ArF amplifier: peak power $P^{beam} = 1\,\text{MW/cm}^3$ and gas pressure $p = 2$ atm. The inset shows the EEDF in the low-energy region. For kinetic energies below about 10 eV and fluorine concentration <1%, the EEDF is close to Maxwell. For $x_{F_2} > 1\,\%$, the low-energy part of the EEDF becomes gradually depleted due to attachment to F_2 molecules and the EEDF swings downward. This observation is consistent with combined Monte Carlo-Boltzmann simulations performed by Kushner,[24] who also observed a downward swing of the EEDF for kinetic energies below about 2 eV. In the inelastic energy region of argon (u > 11.5 eV), the EEDF is not sensitive to fluorine since collisions with Ar atoms dominate the distribution. Figure 1 implies that F_2 can have a significant impact on rates with low-energy threshold (<11.5 eV) such as electron attachment to F_2, while rates with higher energy threshold are mildly affected by F_2 molecules.

The mean electron energy, $\langle u\rangle = \int_0^\infty uf(u)du/\int_0^\infty f(u)du$, and electron density n_e vs. fluorine concentration are plotted in Figs. 2(a) and 2(b), respectively. Here, P^{beam} is varied between 0.01 and 1 MW/cm^3. With increasing fluorine concentration, the mean electron energy increases since F_2 removes electrons from the low-energy part of the EEDF via attachment, a phenomenon broadly known as "attachment heating." Our simulation results and overall trend of $\langle u\rangle$ are consistent with previously published ones.[24,28] For fluorine mole fraction between 0.01% and 1% and low e-beam power deposition, the effective electron temperature $T_e = (2/3)\langle u\rangle$ increases from ~1 eV to ~1.7 eV, which is only 10%–15% larger than Kushner's simulations.[24] At $x_{F_2} \cong 1\%$ the mean energy $\langle u\rangle$ rolls over and decrease. Variation in beam power deposition leads to a weak increase of $\langle u\rangle$ due to the competition between attachment and Coulomb collisions. The electron density scales inversely with the fluorine concentration and increases with power deposition, but in both cases, the scaling is not strictly linear since the dissociative

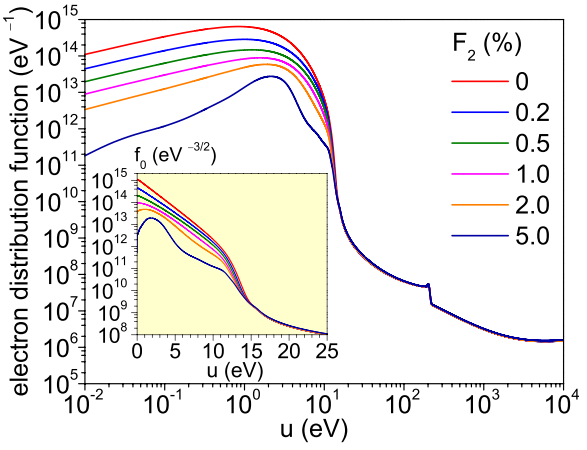


FIG. 1. Calculated EEDF for gas pressure $p = 2$ atm, beam power $P^{beam} = 1\,\text{MW/cm}^3$, and various fluorine concentrations. Inset: low-energy part of the EEDF.

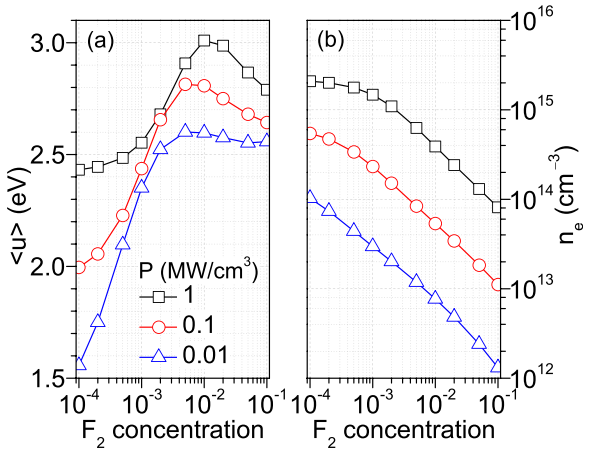


FIG. 2. Calculated mean energy (a) and electron density (b) vs. F_2 concentration for beam powers of 0.01, 0.1, and 1 MW/cm^3. The gas pressure is $p = 2$ atm.

recombination rate scales as electron density squared. The downward swing of n_e indicates that for $x_{F_2} > 0.1\,\%$ attachment to F_2 molecules is the primary loss channel for electrons.

The dominant process driving the plasma is ionization. The total ionization rate is quantified by the energy per electron-ion pair plotted in Fig. 3(a), and the combined individual ionization rates, R_α, are expressed through the γ_α's [Fig. 3(b)]. For $P^{beam} = 1\,MW/cm^3$ and $p = 2$ atm, we calculate $W_{ei}(Ar) = 25.5$ eV and $W_{ei}(F_2) = 28.5$ eV in pure Ar and F_2, respectively. The "averaged" energy per electron-ion pair in an Ar-F_2 mixture is independent of P^{beam} but increases with fluorine concentration since $W_{ei}(F_2) > W_{ei}(Ar)$. The energy per electron-ion pair for argon has been extensively studied and is known to within about 10%, see Refs. 24 and 29 and references therein. Our value, 25.5, is between the ones commonly quoted in the literature (25–27 eV). The normalized ionization coefficients γ_{Ar} and γ_{F_2} follow a similar behavior: they increase with x_{F_2} and are also independent of P^{beam}. For F_2, the combined ionization rate is split between ionization to F_2^+ (83%) and dissociative ionization (17%), both being independent of x_{F_2} and P^{beam}. For Ar, direct ionization accounts for 97.5% and Auger ionization for 2.5% of the combined ionization rate.

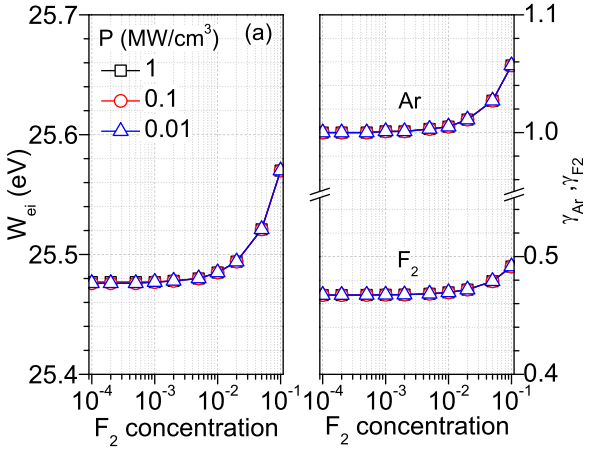


FIG. 3. Calculated energy per-electron-ion pair (a) and coefficients γ (b) vs. F_2 concentration for beam powers of 0.01, 0.1, and 1 MW/cm^3. The gas pressure is p = 2 atm.

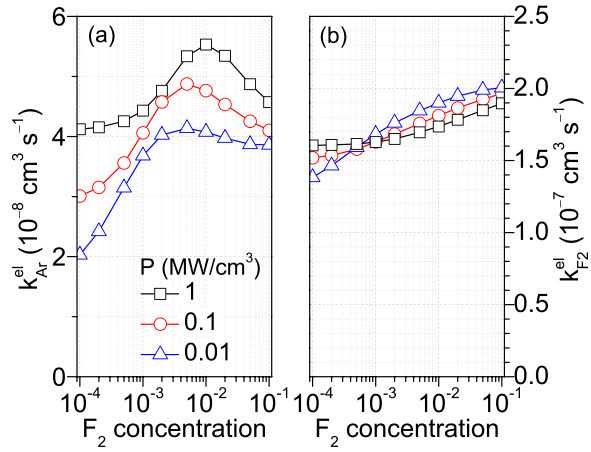


FIG. 4. Calculated elastic scattering rates for Ar (a) and F_2 (b) vs. F_2 concentration for beam powers of 0.01, 0.1, and 1 MW/cm^3. The gas pressure is $p = 2$ atm.

The rate coefficients for elastic scattering of Ar and F_2 vs. x_{F_2} are plotted in Fig. 4. For Ar, the rate coefficient follows closely the trend with $\langle u \rangle$. The excitation-to-ionization ratios for Ar fall in two categories: those for excited states with the lowest energy threshold such as Ar(4s) and Ar(4p) and those with higher energy thresholds. The states in the first category are affected by both *e*-beam power and fluorine concentration, in accordance with the sensitivity of EEDF to fluorine in the energy region between the first excitation energy and ionization energy of Ar [Fig. 5(a)]. The overall trend for η's is to gently decrease with x_{F_2} increasing and level off at a fluorine molecular concentration of ~1%. For the higher argon excited states, Ar(3d), Ar(5s), Ar(4d), Ar(5p), and Ar(6s), the excitation-to-ionization ratios are constant [Figs. 5(b) and 5(c)]. Electron impact excitation of F_2 is treated the same way as Ar, using the excitation-to-ionization ratio, which is plotted in Fig. 5(d).

There are a number of electron collisional processes with F_2 molecules that have low-energy threshold such as vibrational excitation and dissociation to neutral products.

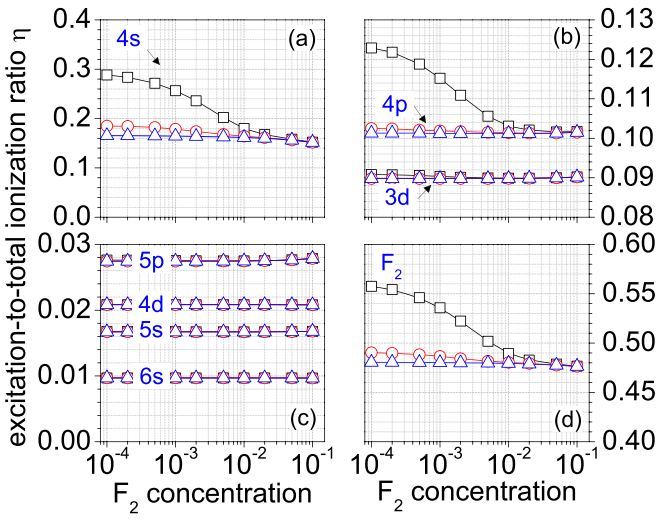


FIG. 5. Calculated excitation-to-ionization ratios for Ar (a)–(c) and F_2 (d) vs. F_2 concentration for beam powers of 0.01, 0.1, and 1 MW/cm^3. The gas pressure is $p = 2$ atm.

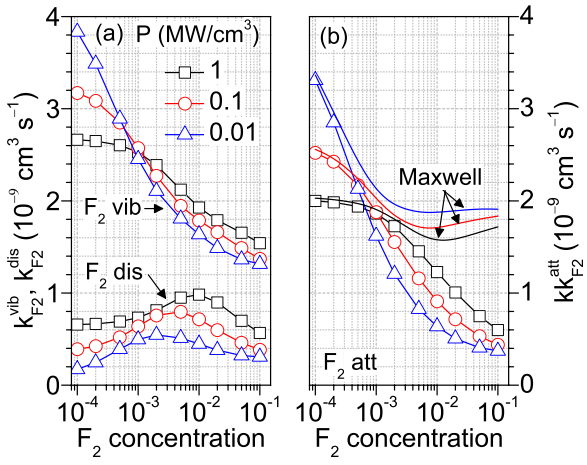


FIG. 6. Calculated vibrational excitation and dissociation rates for F_2 (a) and dissociative attachment rates (b) vs. F_2 concentration for beam powers of 0.01, 0.1, and 1 MW/cm^3. Solid lines: rates calculated with a Maxwell EEDF. The gas pressure is $p = 2$ atm.

The attachment to F_2 molecules has no threshold at all. For these processes, rate coefficients are better suited for approximation than excitation-to-ionization ratios since they are in the bulk of the electron energy distribution. These rate coefficients are shown in Fig. 6. One of the most important processes is dissociative attachment to F_2 molecules. It is the main channel for removing electrons and determines the electron density in the plasma. It is also one of the most sensitive to the EEDF, as conjectured from Fig. 1, since the cross section is the largest at low kinetic energies (<1 eV), where the EEDF is depleted. Though it is a well-known fact that in the noble gas-fluorine mixture the EEDF may not be Maxwell and rate constants obtained by assuming so may not be accurate,[24] it is illuminating to show the effect and quantify the difference specifically for the Ar-F_2 mixture. Figure 6(b) compares rates calculated with the EEDF to those calculated with a Maxwell energy distribution. At very low concentrations, the rates are approximately the same, which is to be expected since the EEDF is not far from a Maxwell distribution. With the increasing F_2 concentration, the rates computed with the EEDF deviate progressively from the rates computed with a Maxwell. At $x_{F_2} = 1\,\%$, the Maxwell distribution overestimates the attachment rates by a factor of two, which would lead to a corresponding error in the electron density determination.

The energy per electron-ion pair W_{ei}, coefficients γ and η for Ar and F_2, and rate coefficients for F_2 have been fitted with appropriate analytical functions, which are summarized in Table I. They are a function of two parameters, the normalized power deposition rate $\bar{P} = \frac{2P^{beam}\,[\mathrm{MW/cm^3}]}{p\,[\mathrm{atm}]} + 10^{-3}$ and F_2 concentration x_{F_2}. For the nominal peak power and pressure ($P^{beam} = 1\,\mathrm{MW/cm^3}$ $p = 2$ atm), the normalized power is $\bar{P} = 1$. The term 10^{-3} has been added to ensure numerical stability at power deposition below 1 kW/cm^3.

IV. FLUORINE BURNUP AND DEPLETION

Large e-beam power deposition combined with minute fluorine concentration (<1%) and large electron density with respect to fluorine density ($n_e/n_{F_2} \geq 10^{-2}$) can lead to fluorine molecular consumption. This is a well-known

TABLE I. Reaction rates R, rate coefficients k, proportionality coefficient for collisional ionization γ defined by Eq. (3), and excitation-to-ionization ratios η defined by Eq. (4). Subscripts denote the species and superscripts denote the collisional process. All rates are spatially averaged and assume uniform deposition across the lasing active media.

Reaction	Rate	Rate coefficient or excitation-to-ionization ratio
Ar combined ionization	$R_{Ar} = \gamma_{Ar} x_{Ar} R^{ion}_{tot}$	$\gamma_{Ar} = 1 + 0.57 x_{F_2}$
F_2 combined ionization	$R_{F_2} = \gamma_{F_2} x_{F_2} R^{ion}_{tot}$	$\gamma_{F_2} = 0.467 + 0.25 x_{F_2}$
$Ar + e \rightarrow Ar^+ + 2e$	$R^{ion}_{Ar} = \eta^{ion}_{Ar} R_{Ar}$	$\eta^{ion}_{Ar} = 0.975$
$Ar + e_{Auger} \rightarrow Ar^+ + 2e$	$R^{Aug}_{Ar} = \eta^{Aug}_{Ar} R_{Ar}$	$\eta^{Aug}_{Ar} = 0.025$
$F_2 + e \rightarrow F_2^+ + 2e$	$R^{ion}_{F_2} = \eta^{ion}_{F_2} R_{F_2}$	$\eta^{ion}_{F_2} = 0.83$
$F_2 + e \rightarrow F^+ + F + 2e$	$R^{di}_{F_2} = \eta^{di}_{F_2} R_{F_2}$	$\eta^{di}_{F_2} = 0.17$
$Ar + e \rightarrow Ar(4s) + e$	$R^{exc}_{Ar,4s} = \eta^{exc}_{Ar,4s} R_{Ar}$	$\eta^{exc}_{Ar,4s} = 0.155 + 0.12 e^{-180 x_{F_2}} \bar{P}^{0.6}$
$Ar + e \rightarrow Ar(4p) + e$	$R^{exc}_{Ar,4p} = \eta^{exc}_{Ar,4p} R_{Ar}$	$\eta^{exc}_{Ar,4p} = 0.101 + 0.023 e^{-400 x_{F_2}} \bar{P}^{1.2}$
$Ar + e \rightarrow Ar(3d) + e$	$R^{exc}_{Ar,3d} = \eta^{exc}_{Ar,3d} R_{Ar}$	$\eta^{exc}_{Ar,3d} = 0.0898 + 0.004 x_{F_2} + 0.001 e^{-600 x_{F_2}} \bar{P}^{1.5}$
$Ar + e \rightarrow Ar(5s) + e$	$R^{exc}_{Ar,5s} = \eta^{exc}_{Ar,5s} R_{Ar}$	$\eta^{exc}_{Ar,5s} = 0.0167$
$Ar + e \rightarrow Ar(5p) + e$	$R^{exc}_{Ar,5p} = \eta^{exc}_{Ar,5p} R_{Ar}$	$\eta^{exc}_{Ar,5p} = 0.02745 + 0.004 x_{F_2} + 0.0002 e^{-600 x_{F_2}} \bar{P}^{1.5}$
$Ar + e \rightarrow Ar(4d) + e$	$R^{exc}_{Ar,4d} = \eta^{exc}_{Ar,4d} R_{Ar}$	$\eta^{exc}_{Ar,4d} = 0.0208$
$F_2 + e \rightarrow F_2^* + e$	$R^{exc}_{F_2} = \eta^{exc}_{F_2} R_{F_2}$	$\eta^{exc}_{F_2} = 0.48 + 0.075 e^{-250 x_{F_2}} \bar{P}^{0.9}$
$Ar + e \rightarrow Ar + e$	$R^{el}_{Ar} = k^{el}_{Ar} n_e n_{Ar}$	$k^{el}_{Ar} = \frac{a}{1+(b+c x_{F_2})^{-1}+d\sqrt{x_{F_2}}}$, $a = (3.9 + 2.2\bar{P}^{0.3}) \times 10^{-8}$, $b = 0.5 + 1.5\bar{P}$, $c = 4.1 \times 10^3/(1 + 1.7\bar{P})$, $d = \bar{P}^{0.09}$
$F_2 + e \rightarrow F_2 + e$	$R^{el}_{F_2} = k^{el}_{F_2} n_e n_{F_2}$	$k^{el}_{F_2} = \frac{a}{1+(b+c x_{F_2})^{-1}}$, $a = (1.92 + 0.1 e^{-10\bar{P}}) \times 10^{-7}$, $b = 1 + 4\bar{P}^{0.25}$, $c = 1.6 \times 10^3/(1 + 3\bar{P})$
$F_2 + e \rightarrow F^- + F$	$R^{att}_{F_2} = k^{att}_{F_2} n_e n_{F_2}$	$k^{att}_{F_2} = a + \frac{b}{1+c\sqrt{x_{F_2}}}$, $a = 0$, $b = (2.3 - 0.5\ln(\bar{P})) \times 10^{-9}$, $c = 50/(1 + 4\bar{P})$
$F_2 + e \rightarrow F_2(v) + e$	$R^{vib}_{F_2} = k^{vib}_{F_2} n_e n_{F_2}$	$k^{vib}_{F_2} = a + \frac{b}{1+c\sqrt{x_{F_2}}}$, $a = 1 \times 10^{-9}$, $b = (2 - 0.45\ln(\bar{P})) \times 10^{-9}$, $c = 50/(1 + 4\bar{P})$
$F_2 + e \rightarrow 2F + e$	$R^{dis}_{F_2} = k^{dis}_{F_2} n_e n_{F_2}$	$k^{dis}_{F_2} = \frac{a}{1+(b+c x_{F_2})^{-1}+d\sqrt{x_{F_2}}}$, $a = (0.7 + 1.7\bar{P}^{0.35}) \times 10^{-9}$, $b = 0.2 + 0.15\bar{P}^{0.4}$, $c = 1 \times 10^3/(1 + 4\bar{P})$, $d = 10$

phenomenon, discovered in the 1970s.[15,30–34] In our model, the two leading reactions causing fluorine burn-up are

$$F_2 + e \rightarrow F^- + F, \quad k^{att} \cong 1.5 \times 10^{-9}\,\mathrm{cm^3\,s^{-1}}, \qquad (5)$$

$$F_2 + e \rightarrow 2F + e, \quad k^{dis} \cong 1.0 \times 10^{-9}\,\mathrm{cm^3\,s^{-1}}, \qquad (6)$$

with rate coefficients taken for $P^{beam} = 1\,\mathrm{MW/cm^3}$, $p = 2\,\mathrm{atm}$, and $x_{F_2} = 0.4\,\%$ (Fig. 6). Nighan[32] considered a wider variety of processes and found that the production of ArF* via the reaction $Ar^* + F_2 \rightarrow ArF^* + F$ can contribute significantly (28%) with direct electron impact dissociation (with a small admixture of Kr ~5%), Eq. (6), being the leading mechanism for fluorine molecule destruction. Accounting for only Eqs. (5) and (6), the number of fluorine molecules decays exponentially, $n_{F_2}(t) = n_{F_2}(0)e^{-t/\tau}$, on a time scale $\tau = [(k^{att} + k^{dis})n_e]^{-1} \simeq 400$ ns evaluated for peak electron density $n_e \cong 10^{15}$ cm^{-3} [Fig. 2(b)]. This time scale is about twice the *Electra*'s e-beam pulse duration, which indicates that fluorine burnup during e-beam deposition can be an issue. Qualitative estimates for the consumption rate and leading parameter determining that the rate can be obtained by solving the rate balance equation for F_2 molecules under the following assumptions: (i) the electron loss in Eq. (1) is dominated by attachment, Eq. (5), (ii) the dissociation to neutral products, Eq. (6), contributes an additional ~60%; and (iii) re-association of F atoms to F_2 proceeds on a time scale much longer than the *e*-beam pulse duration. Since $R^{att} \cong R^{ion}_{tot} = P^{beam}/(W_{ei}e_0)$ and $R^{dis} \cong 0.6R^{att}$, the particle balance equation for F_2 molecules takes the form

$$\frac{dn_{F_2}(t)}{dt} \cong -1.6P^{beam}(t)/(W_{ei}e_0). \qquad (7)$$

Equation (7) is readily integrated over time to yield the number of dissociated molecules per unit volume for the duration of the *e*-beam pulse

$$\Delta n_{F_2} \cong 1.6\varepsilon^{beam}/(W_{ei}e_0). \qquad (8)$$

To a first order approximation, the rate of fluorine molecule burn-up depends only on the *e*-beam energy deposition per unit volume, $\varepsilon^{beam} = \int_0^\infty P^{beam}(t)dt$. Setting $W_{ei} \cong 25\,eV$, Eq. (8) takes final form: $\Delta n_{F_2}[\mathrm{molecules/cm^3}] \cong 4 \times 10^{17}\,\varepsilon^{beam}[\mathrm{J/cm^3}]$. This is in excellent agreement with experimental data,[34] as seen in Fig. 7. The effect of molecule consumption can be counteracted by increasing the filling F_2 pressure above the minimal desired concentration with a number of F_2 molecules per unit volume according to

$$(n_{F_2})_{input} \cong (n_{F_2})_{\min} + 4 \times 10^{17}\varepsilon^{beam}. \qquad (9)$$

A consideration in designing an ArF *e*-beam angularly mutiplexal laser system is the index of refraction of the gain media, which impacts the focusing and defocusing in the gain media.[35] The electron beam pumping of ArF lasers creates an additional dynamic change during the *e*-beam pulse to the index of refraction. The index of refraction change in ArF lasers, similar to other electron beam excimer lasers,[36] is composed of two interrelated parts: electron density and

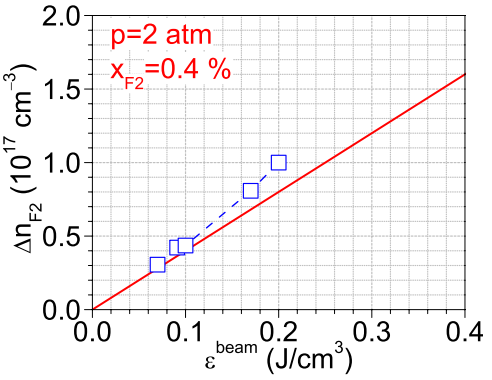


FIG. 7. Number of dissociated F_2 molecules per unit volume vs. beam energy deposition rate for $P^{beam} = 1\,\mathrm{MW/cm^3}$, $p = 2\,\mathrm{atm}$, and $x_{F_2} = 0.4\,\%$. Symbols denote the experiment,[34] and solid lines denote the model.

fluorine burnup. Both of these changes are dynamic during the duration of the electron beam pulse and compose the transient refractive index. Both components can be estimated based on the results from Secs. III and IV with knowledge of the refractivity of F and F_2, which are $\bar{n}_{0,F_2} = 1.000088$ and $\bar{n}_{0,F} = 1.000195$ per amagat, respectively.[36] The change of the index of refraction for the electron density, $(\Delta n)_{electrons}$, is calculated with the plasma frequency ($\omega_p = \sqrt{4\pi n_e e^2/m_e}$) and laser frequency ($\omega$) according to

$$(\Delta n)_{electrons} = \left(\sqrt{1 - \omega_p^2/\omega^2}\right)_{t=0} - \left(\sqrt{1 - \omega_p^2/\omega^2}\right)_{t=t_{end}}. \qquad (10)$$

The change in index of refraction due to fluorine burn up, $(\Delta n)_{F_2}$, is calculated with the change in molecular fluorine number density. In the simplest case of molecular fluorine dissociation ($F_2 \rightarrow 2F$)

$$(\Delta n)_{F_2} = \frac{n_{F_2}(t=0) - n_{F_2}(t=t_{end})}{n_0}\left(2\beta_F - \beta_{F_2}\right), \qquad (11)$$

where $\beta_\alpha = \bar{n}_{0,\alpha} - 1$ ($\alpha = F, F_2$) and n_0 is the Loschmidt number. The total change $(\Delta n)_{tot}$ is the summation of the two components. For the case of $P^{beam} = 1\,\mathrm{MW/cm^3}$ and pulse length of 150 ns in a square pulse, the fluorine burnup is the larger contributor as shown in Table II. The estimates

TABLE II. Results of the transient refractive index for the ArF e-beam amplifier with $P^{beam} = 1\,\mathrm{MW/cm^3}$ and $\tau^{beam} = 150$ ns square pulse as a function of F_2 concentration. Time $t = 0$ is the onset of the electron beam pulse and $t = t_{end}$ is the turn off of the electron beam pulse. The symbols $(\Delta n)_{electrons}$, $(\Delta n)_{F2}$, and $(\Delta n)_{tot}$ are the change in the refractive index from onset of e-beam pulse to the end of e-beam pulse for electron density, change in fluorine density, and total refractive index change, respectively.

x_{F_2} (%)	0.15	0.3	0.5	1
$n_{F_2}(t=0)$ ($\times 10^{17}$ cm^{-3})	0.75	1.5	2.5	5
$n_{F_2}(t=t_{end})$ ($\times 10^{17}$ cm^{-3})	0.15	0.9	1.9	4.4
n_e ($t=0$) ($\times 10^{14}$ cm^{-3})	13	9.9	6.3	3.9
n_e ($t=t_{end}$) ($\times 10^{14}$ cm^{-3})	19	12	8.8	4.4
$(\Delta n)_{electrons}$ ($\times 10^{-9}$)	−10	−5.1	−4.1	−0.8
$(\Delta n)_{F2}$ ($\times 10^{-9}$)	−42	−42	−42	−42
$(\Delta n)_{tot}$ ($\times 10^{-9}$)	−52	−47	−46	−43

indicate that the total refractive index change is very small and its impact will be negligible in ArF laser design.

V. SUMMARY AND CONCLUSIONS

In conclusion, a detailed study of the electron energy deposition relevant to the *e*-beam pumped ArF laser amplifier has been performed. Collisional rates of background atoms and molecules with electrons have been calculated as a function of reduced *e*-beam power density and fluorine concentration. In order to ensure sufficient accuracy, the rates must be calculated with the EEDF instead of Maxwell distribution. The dissociative attachment rate is particularly sensitive to the EEDF, and using a Maxwell can lead to an error of a factor of two. The burnup of F_2 molecules has been investigated, and a simple formula has been proposed to account for that effect in designing the initial F_2 concentration. The expected transient refractive index for ArF electron beam pumped lasers is very small.

In future work, we will employ these rates into a model of the ArF laser, which includes a 1D time-dependent plasma chemistry code coupled to a 3D radiation transport for the amplified spontaneous emission.

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